



MIT's heart rate and breathing monitor at work

sleep a little longer if you arrived late from office the previous night and do not have a meeting till noon that day.

Another interesting scenario detailed in the patent is that of multiple homes connected together to form a smart neighbourhood that will have improved security, cooperation for energy management, the ability to alert one another in case of events like fire and accidents and so on.

This door has no handle or lock to break

Mitsubishi Electric's model smart-home in Tokyo shows off some wonderful futuristic technologies, starting right from the door handle. When you arrive at the house, all you can see in front of you is a snazzy metallic sheet with a small peephole. There is no handle or lock visible on the door. But as soon as the biometric camera (yes, what you thought was a peephole) has scanned you and recognises you as an authorised inhabitant, a door handle pops out of the sheet.

As soon as you touch the handle to pull, it quickly and unobtrusively checks your temperature and heart rate to ensure you are okay. If there are any updates for you, you will get those right away—it will tell you if your parcel has arrived or your child is yet to return from school and so on. You can also check the wherea-

bouts of other family members and when they are expected home (as the door syncs to their schedules, too).

If that sounds too futuristic, let us step into the kitchen. But before you do that, your vital statistics and that of your family members have already reached your fridge. Based on the stock inside the fridge and depending on the health parameters collected by the door handle and other connected devices being used by the rest of your family, this smartdevice gives you suggestions on what to cook.

Once you decide on the menu, step-by-step cooking instructions and estimated preparation times are displayed on the kitchen countertop. The countertop has movable, ultra-thin induction heaters that can slide from end-to-end, as well as outlet-free, non-contact power for blenders, steamers and other kitchen appliances.

According to news reports, the company has no immediate plans to launch these devices. It wants to take time to design the products carefully so that these can gel with users' lives, to help without overwhelming them.

A drone to guard your house

This year's CES saw interesting drones being demonstrated by companies like Yuneec, Hexo+ and Hobbyco. In a post-event interview, Ryan

Kephart of Yuneec suggested that drones for home surveillance were a practical but unexplored area.

Drones are stealthy; these can hover about and can be controlled by smartphones. Today's drones are capable of continuously streaming audio and video, too. So why cannot a combination of drones and software be used to watch a house? This could be an interesting twist to future smarthomes.

Wearables that turn into authenticators

In a *Wired* article, Kevin Foreman, director of product vision at Vectorform, explains that, wearables will play a significant role in the future smarthome. These would be like your key to the smarthome.

Nymi, for example, is a wristband that continuously monitors your heartbeat and uses it as your signature to authenticate you to several other systems. It eliminates the overhead of having to manually identify yourself to other systems and devices using passwords or fingerprints. Plus, your heartbeat is inimitable, so it is a very secure technique, too.

Over time, Nymi could also be improved to track the user's body temperature to adjust air-conditioning, accordingly.

Smarthome tech for a healthy heart

While many people think of smart-homes only from the perspective of energy management, comfort and entertainment, among other things, there are some for whom a smart-home could be a life-support system.

Here is where MIT's new wireless sensing technology, Vital-Radio, can be of help. The device, when built into the walls of a home, can monitor the occupants' heart rate and breathing without requiring them to wear even a single sensor.

According to the results published in a research paper, the technology boasts a median accuracy of